

ASSESSMENT BRIEF: SOLUTION ARCHITECTURE

MODULE CODE	UA92- 231
MODULE TITLE	Solution Architecture
MODULE LEADER	Abdullahi Hasan, Iqra Kiyani
CREDIT VLAUE	25
ASSESSMENT TYPES & WEIGHTING	Pre-Recorded Presentation (30%) Report (70%)
SUBMISSION FORMAT	Pre-Recorded Presentation: MP4 file/link & .pptx file & MS Forms Report: Word Document (.docx) and ZIP file
SUBMISSION DEADLINE	18.00 on 31/10/2025 Submit on Teams Assignment area

ASSESSMENT LEARNING OUTCOMES

On successful completion of this module you will be able to:

1. Use an Architecture tool to design and define a solution.
2. Locate appropriate existing (well-tested modular) components.
3. Define the solution, integrating/synthesising the components in readiness for the system to be developed.
4. Realise and manage the risk and security implications.
5. Extract information from non-technologically competent stakeholders.
6. Communicate the solution in an appropriate format to individuals with a range of knowledge (in both the business and the technical arena).

ASSESSMENT SCENARIO

You are working as a Solution Architect for the Uzumaki Arts Academy, a distinguished place dedicated to the study and practice of the arts. The institute is at an exciting juncture where it seeks to blend traditional creative disciplines with contemporary technology. As the institute expands, the need for a centralised system to handle its varied operations has become more evident.

This has led to a pressing requirement for a bespoke Customer Relationship Management (CRM) system. The aims of this system are to streamline business processes, reduce costs, and foster a smooth and efficient environment for all stakeholders. The main challenge is to ensure that the CRM respects the unique nuances of the arts while meeting the business requirements.

Your task is to plan and design the architecture for this CRM system, ensuring it can store, process, and analyse a diverse range of data and variables specific to the institute's activities. You are not expected to implement or programme the CRM itself, your focus is solely on its design.

ASSESSMENT 1: Requirements Capture and a Picture in Pre-recorded Presentation [LO 3, 5 & 6]

WEIGHTING: 30%

TIMING: 10 minute presentation $\pm 10\%$

SUBMISSION FORMAT: MP4 and/or link to the video held on Stream (if recorded via Teams) & . pptx file, & MS Forms questionnaire

Task - Requirements Capture

1. Devise a concise, targeted questionnaire using domain-specific language to gather key information from the admissions manager about the admissions department, with particular emphasis on the architecture of the supporting IT system.
2. Justify the relevance and appropriateness of the questions included, providing detailed reasoning for your choices through a recorded presentation and supporting documentation.
3. Evaluate the effectiveness of the chosen requirements elicitation method by role-playing as the admissions manager, generating realistic user requirements to inform Assessment 2, clearly explaining the rationale behind each answer provided.

SUBMISSION DETAILS

RELEASE DATE	22/09/2025
SUBMISSION DATE	18.00 on 09/10/2025
DELIVERABLES	Submit the following: - Questionnaire - Create clear and succinct questions targeted at the admissions manager. - Focus specifically on the admissions department and its supporting IT system. - Frame questions within the context of architecture, avoiding technical jargon and programming terminology. - Ensure domain-specific terminology appropriate for a non-technical business user. AI : Partially Allowed (see AI RAG rating below) - Presentation - Prepare a 10-minute pre-recorded presentation, scheduled for Week 3 - Assume the role of the admissions manager. - Answer the devised questionnaire, generating realistic user requirements. - Clearly discuss and evaluate your chosen requirements elicitation method. - Justify the selection of questions in your questionnaire, explaining their relevance and depth. - Provide a rationale for answers given, demonstrating how they translate into system requirements. AI (RED: AI use not permitted):
SUBMISSION DETAILS	Submit your assignment by uploading it to the Teams assignment area by 18:00 PM on the submission date. This deadline will be automatically and strictly enforced. If your submission is late, your grade may be affected. If you have any issues submitting your work, you must message your tutor <u>before the assessment due date</u>. Do not leave your submission until the last minute to avoid any penalties due to problems with the submission.

	We may ask for a demonstration of your work following the submission. If needed, this will be communicated to you individually.
MARK & FEEDBACK	Your mark and feedback for this assessment will be provided within 21 working days.

UA92 MARKING SCHEME

Classification	Mark Range
Exceptional 1st	100
Very High 1st	92
High 1st	85
1st (First Class)	78 / 72
2:i (Upper Second)	68 / 65 / 62
2:ii (Lower Second)	58 / 55 / 52
3rd (Third Class)	48 / 45 / 42
Fail	35 / 25 / 15 / 0

ASSESSMENT 1 MARKING CRITERIA

Fail (0 – 39%)	- The Questionnaire is unclear, includes irrelevant questions and lacks justifications for their selection. - The questionnaire shows little or no focus on admission department - Either no presentation submitted or it lacks clear structure, little or inadequate content, and poor delivery.
To obtain a 3 rd mark (40-49%), you must have:	- The Questionnaire contains several unclear or partially irrelevant questions, with limited justifications for their selection. - The questionnaire shows a basic attempt to focus on admission department but lacks in depth analysis. - The Presentation demonstrates some structure but has limited content, weak logical flow quality is poor, with minimal content, weak structure, and inconsistent delivery.
To obtain a 2:ii mark (50-59%), you must have (in addition to the above):	- The Questionnaire include most relevant questions, though some lacks clarity and full justification. - The questionnaire shows a general focus on admission processes and within the context of architecture, while avoiding technical jargon and programming terminology. - Presentation has adequate content and structure, though greater clarity and effective delivery are needed.
To obtain a 2:i mark (60-69%), you must have (in addition to the above):	- The questionnaire is clear, relevant, and most well justified the selection of questions. - The questionnaire shows a strong focus on admission processes, the supporting IT systems, and their relevance to solution architecture, - The presentation is well-structured, coherent, and delivers content clearly, with minor areas for improvement.
To obtain a 1st mark	The questionnaire is comprehensive, clear, high relevant,

(70-100%), you must have (in addition to the above):	and fully justified the selection of questions. • The questionnaire demonstrates an excellent understanding admission processes and their supporting IT systems within a well-defined solution architecture context. • Communicates complex ideas clearly and accessible to non-technical audience. • The presentation is highly structured, engaging, well-delivered, and shows depth of content and analysis.
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ASSESSMENT 2: Solution Architecture Design [LO 1,2,3 & 4]

WEIGHTING: 70%

EQUIVALENT WORD COUNT: 3000 ±10%

SUBMISSION FORMAT: Word Document (.docx) + Architecture Designs (ZIP)

Background - The current system

The admissions team have several sub-divisions, and the current admissions system (below) has been identified as one that is no longer fit for the purpose:

1. IT division (made up of in-house first-level IT support staff)
2. Administration division (made up of admissions staff)
3. Research division (made up of academic administration staff)
4. Communications division (made up of student experience staff)
5. Decision-making division (made up of academic staff)
6. Management division (made up of 3 admissions managers)

The admissions manager is satisfied with your requirements (from Assessment 1) and your understanding of the current system. Thus, you have been given the task of designing the architecture for the new admissions CRM system. You do **NOT** need to create the new system, you are only to design the system architecture.

RELEASE DATE	22/09/2025
SUBMISSION DATE	18.00 on 31/10/2025
DELIVERABLES	Task 1 – Enterprise Architecture Frameworks • A detailed evaluation of four common architecture frameworks: ◦ Zachman Framework ◦ Federal Enterprise Architecture Framework (FEAF) ◦ The Open Group Architecture Framework (TOGAF) ◦ Department of Defense Architecture Framework (DoDAF) • A justified selection of the most suitable framework for the admissions system, clearly explaining why this framework aligns best with the solution’s needs. • An analysis of alternative frameworks considered, including explicit reasons for their rejection, supported by evidence. AI AMBER partially permitted Task 2 – Architecture Modelling and CRM Solution • A completed Architectural Decision Document using the selected framework’s template. • Clearly stated assumptions made in the design of the CRM solution for Uzumaki Arts College’s admissions department. • An explicit consideration of all elements within the chosen architecture framework, identifying and justifying any elements deemed ‘out of scope’. • Reference and justification provided if the solution is modelled on an existing system. • Appropriate architectural diagrams created using tools such as ArchiMate and diagrams.io, illustrating system inputs, outputs, and interactions clearly. • GAP analysis clearly integrated into the solution, highlighting areas for improvement or further development. AI RED not allowed

	Task 3 – System Security Documentation • A formal document assessing architectural risks and security implications of the proposed CRM solution. • A detailed consideration of both current and future security risks, framed within the selected architecture framework context. • Clear focus and documentation of compliance issues related to regulatory and data protection requirements relevant to the admissions system. AI RED not allowed Documents must have a coherent structure supported by thorough research and referencing. Work must be referenced using the Harvard referencing system. Please use Arial Font 12pt (14 pt for Headings) and 1.5 spaced. Naming Convention: ID_NAME_Assessment Number For Example, 1234343_SaraSmith_Assessment 1
SUBMISSION DETAILS	Submit your assignment by uploading it to the Teams assignment area by 18:00 PM on the submission date. This deadline will be automatically and strictly enforced. If your submission is late, your grade may be affected. If you have any issues submitting your work, you must message your tutor <u>before the assessment due date</u>. Do not leave your submission until the last minute to avoid any penalties due to problems with the submission.
	We may ask for a demonstration of your work following the submission. If needed, this will be communicated to you individually.
MARK & FEEDBACK	Your mark and feedback for this assessment will be provided within 21 working days.

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ASSESSMENT 2 – MARKING CRITERIA

Fail mark (0 – 39%)	- Framework evaluations are missing or incorrect, with no justification for the chosen framework or analysis alternatives. - The Architecture decision document is missing, or largely incomplete, key framework elements are omitted or misunderstood. - Assumptions for the CRM solution are missing or irrelevant. - No evidence of system modelling (diagrams), gap analysis, or alignment to an existing system. - Documentation on system security and regularity compliance is unclear, lacks structure, contains little or no risk analysis
To obtain a 3 rd mark (40%), you must have:	- Framework evaluations are attempted but contain major inaccuracies, and the justification for the chosen framework or analysis alternatives is weak or incomplete. - The Architecture Decision Document is incomplete, with several key elements missing only briefly addressed. - CRM assumptions are minimal, vague, or partially relevant. - Basic attempt of system modelling (diagrams), gap analysis, or alignment to an existing system.

	Documentation on security and regulatory compliance is poorly structured and lacks in depth analysis, with very limited risk analysis.
To obtain a 2:ii mark (50%), you must have (in addition to the above):	- Framework evaluations are mostly correct but lacks through justification for the chosen framework, with little consideration of alternatives. - The Architecture Decision Document covers most required elements, though some lacks detailed accurate description. - CRM solution assumptions are stated and mostly relevant but may be underdeveloped. - System modelling and gap analysis included, with some attempts to align an existing system. - Documentation on security and regulatory compliance is generally clear but only includes basic risk analysis.
To obtain a 2:i mark (60%), you must have (in addition to the above):	- Framework evaluations are accurate, demonstrate a clear understanding, and provide well-reasoned justification for the chosen framework as well as the alternatives included in comparative analysis. - The Architecture Decision Document is complete and coherent, covering all required elements with reasonable clarity. - CRM solution assumptions are clearly stated and mostly well justified. - System modelling and gap analysis are clear and logical, showing good alignment of an existing system. - Documentation on security and regulatory compliance is well structured, with clear risk assessment, though may lack some depth analysis in some parts.
To obtain a 1st mark (70%), you must have (in addition to the above):	- Framework evaluations are comprehensive and accurate, providing well-justified reasoning for the chosen framework along with a clear comparative analysis. - The Architecture Decision Document is complete, well structured, and demonstrates deep understanding of all required framework elements. - CRM solution assumptions are clear, and fully justified, showing critical thinking. - System modelling and gap analysis are detailed and clearly aligned to existing systems. - Documentation on security and regulatory compliance is thorough, clearly structured, and includes robust risk analysis and mitigation strategies.

ACADEMIC MISCONDUCT

Academic Misconduct includes all forms of academic dishonesty, whether intentional or accidental, that compromise the integrity of the University's assessment processes. It is essential that you review our [Assessment Policy](#) to understand the guidelines and the serious consequences that may arise if they are not followed.

AI Use RAG Guidance

IMPORTANT: Use of AI should be for learning purposes only. Any abuse of this tool will result in Academic Misconduct.

Zone	Example	Permitted?
RED	AI is to prepare pre-recorded presentation. AI is used to evaluate Enterprise Architecture Frameworks. AI is used for Architecture Modelling and CRM Solution. AI is used for System Security Documentation.	No (Use will be treated as Academic Misconduct)
AMBER	AI is used to understand better how to develop a concise, targeted questionnaire using domain-specific language to gather key information from the admissions manager about the admissions department. AI is used to help or suggestion to create appropriate architectural diagrams using tools such as ArchiMate and diagrams.io.	Yes (But must be reviewed, adapted, and declared in report/appendix where AI was used)
GREEN	AI is used for tutorial, revision and to understand and compare four common architecture frameworks: Zachman Framework, Federal Enterprise Architecture Framework (FEAF), The Open Group Architecture Framework (TOGAF), Department of Defense Architecture Framework (DoDAF)	Yes

IMPORTANT MODULE MILESTONES

Week Number	Activities essential to the assessment:
Week One	• Introduction to Block 1 Solution architecture. This session covers the block handbook, assessment details and learning objectives. • Introduce the role of the Solution Architect. • Detailed study of the framework tools. • Enterprise Architecture and its importance. • Dynamic Learning: Introduction to frameworks: TOGAF, Zachman, ArchiMate. Detailed study of TOGAF: investigation of case studies.
Week Two	• The role of solution architecture in a business context and its significance • How the business process works • How business needs are fulfilled • How business objectives are achieved • How solutions deliver value to businesses • 92 Programme sessions • Dynamic Learning: Enterprise Architecture detailed case study work through
Week Three	• BDAT (Business, Data, Applications, Technical) layers of Solution Architecture. • Problem (requirements: technical; business) • 92 Programme sessions • Dynamic Learning: Development methodologies (Agile, Scrum, Lean). • Assessment 1 deadline – ppt presentation
Week Four	• Constraints (resources, compliance etc.) • Options (bespoke: in-house development, off-the-shelf, buy in some/all combinations). • Dynamic Learning: Azure Solution Architecture part1 • 92 Programme sessions
Week Five	• Introduce commonly used Architecture tools and modelling representations, ERD, UML, Visio/diagrams.net, BPM/BPMN. • Project governance and management methodologies: risk analysis, planning and control. • Dynamic Learning: Azure Solution Architecture part2
Week Six	Project work with cohort wide assistance provided. Teaching in week six focusses on the final coursework, assessment, and support for the final assessment work, making amendments plus providing overall support for final submission.

LINKS TO 92 Programme

As part of the 92 Programme (92P), you will focus on key traits and skills which UA92 graduates are expected to be able to demonstrate. These will be developed explicitly through your 92P sessions, but also throughout your learning on the module. So you should consider these skills as you learn in your subject.

Your 92P learning journal is open to you throughout the year to record your progress and reflection, leading to your 92P assessment.

ASSESSMENT REGULATIONS

- The standard University assessment regulations apply for this assessment.
- Please note that failure to submit work (i.e. non-submission) could lead to you failing the module.
- Details of assessment regulations are available at:
- [UA92-Academic-Regs-MARP-24-25.pdf](#)